

E-COMMERCE PROFITABILITY ANALYSIS REPORT

Strategic Diagnostic on Margins, Channels, and Marketing ROI

Prepared by: Paul Tonwe

Date: June 2026

1. INTRODUCTION AND STAKEHOLDER OBJECTIVES

While top-line gross revenue across the enterprise appears structurally healthy, corporate leadership suspected hidden inefficiencies across our product categories and digital sales channels. Operational expenses including shipping friction, unexpected transaction fees, cross-channel platform fees, and marketing capital distributions threatened to erode net margins. This report details a comprehensive data diagnostics project built to isolate hidden profit leaks, optimize channel efficiency, evaluate advertising performance, and establish an actionable data-driven optimization strategy for the Chief Executive Officer.

2. DATA AND METHODOLOGY

The underlying analysis was executed entirely within Power BI Desktop to build an integrated ETL pipeline, analytical schema, and reporting application. The raw data infrastructure consisted of three independent transactional logs: transactional orders (2,000 unique records containing detailed cost and revenue variables), marketing expenditures (monthly platform performance statistics), and a master product inventory catalog.

The Core Metrics Framework

To protect calculation efficiency, all business metrics were engineered using explicit DAX columnar aggregations and conditional division functions rather than row-by-row calculated columns. Key definitions implemented include:

- Total Net Revenue: Summed gross revenue minus explicit customer discounts and promotional impacts.
- Operating Margin: Total net profit divided by total net revenue, establishing a baseline of true profitability per dollar earned.
- Return on Ad Spend (ROAS): Total revenue directly attributed to ad platforms divided by total advertising cash spend.
- Customer Acquisition Cost (CPA): Total advertising spend divided by total platform conversions, reflecting marketing efficiency.

Architectural Choice: The Galaxy Schema Constellation

A standard Star Schema structures data around a single central fact table wrapped in descriptive dimension lookup tables. However, this enterprise architecture presented a significant granularity mismatch: the orders log tracks individual daily transactions, while the marketing expenditure log records data aggregated at a monthly, platform-specific level. Attempting to force these two independent datasets into a single fact table or establishing a direct relationship on text fields would create a destructive Many-to-Many loop, resulting in duplicated values, data bloat, and highly distorted financial calculations.

To eliminate this operational hazard, a Galaxy Schema (Fact Constellation) was architected. This model deploys two independent fact tables (Orders and Marketing Spend) that exist side-by-side. They remain completely unlinked to each other but are connected downstream to shared dimension tables. The master product table was retained as an unlinked analytical reference, preserving model performance while leaving it accessible for independent, catalog-specific side queries.

Relationship Modeling: The Shared Calendar Table Bridge

To bridge the two fact tables at their differing temporal granularities, a custom, continuous Dim_Calendar table was generated. Relationships were constructed using a clean 1-to-Many single-direction filter flow:

- Dim_Calendar[Date] maps directly to the transactional daily grain of fact_orders[order_date].
- Dim_Calendar[Month-Year] maps directly to the aggregated monthly grain of fact_marketing_spend[month].

This centralized date bridge allows an executive user to apply a single temporal filter on a slicer and watch both independent transaction streams calculate in perfect structural harmony, preventing any double-counting of costs.

3. KEY FINDINGS

Product Category Profitability Diagnostics

The store-wide portfolio generated \$277,969.13 in Gross Revenue, yielding \$56,333.83 in Net Profit with a baseline Net Profit Margin of 23.84%. Sorting performance by absolute profit isolates clear performance differences across categories:

- Electronics emerged as the primary portfolio driver, generating \$13,973.46 in profit with a dominant 31.13% margin. This efficiency is driven by highly optimized fulfillment logistics; shipping costs consume just 13.50% of gross revenue, the lowest across the brand.
- Books sits at the bottom of the portfolio, returning a weak 11.94% margin (\$2,249.79 total profit). This underperformance is caused by logistics friction, where shipping costs consume a massive 27.94% of every gross dollar earned.
- Beauty represents a manufacturing margin risk; while it has a low return rate, raw product costs consume a portfolio-high 41.16% of its gross revenue, capping its upside to a 17.39% profit margin.

Sales Channel Efficiency Analysis

Analysis across our four digital sales channels reveals that top-line volume does not always equal bottom-line efficiency once platform costs are factored in:

- The Mobile App is our most efficient transactional engine, generating an Average Profit Per Order of \$36.32 with zero platform fee drag.

- The Website delivers the highest absolute profit (\$25,118.46 across 795 orders) with an Average Profit Per Order of \$31.60.
- Marketplace volume is heavily restricted by structural platform fees, which drained \$7,949.69 in cash from our revenue. This fee drag reduced its Average Profit Per Order to just \$15.40.
- Social Commerce underperformed across all efficiency markers, generating a weak \$17.11 average profit per order due to an elevated 9.14% return rate.

4. CONCLUSIONS AND STRATEGIC RECOMMENDATIONS

Marketing Budget Optimization: 20% Reduction Strategy

The company-wide advertising portfolio represents a total capital allocation of \$503,506.14. Executing the requested 20% budget reduction requires a target cost saving of \$100,701.23. Rather than making a generic cut across all platforms, a targeted reduction preserves high-performing engines while eliminating capital drag:

1. Complete Elimination of Email Marketing: This channel delivered an underperforming 4.81x ROAS with an expensive CPA of \$17.86. Completely cutting this channel removes \$24,461.37 in capital drag.
2. Targeted Reductions in Facebook Ads: Facebook represented our second least efficient channel, with a high CPA of \$6.27 and an 11.45x ROAS. It should be trimmed by approximately \$76,239.86, specifically targeting historical underperforming months like August and September.

This targeted strategy achieves the exact \$100,701.23 savings target while completely protecting our highest-performing assets, including TikTok Ads (24.02x ROAS) and Influencer campaigns (22.70x ROAS).

Dual-Perspective Return Rate Strategy

The total revenue lost to returns across the entire analysis period reached \$20,582.45, with a store-wide baseline return rate of 7.20%. To optimize cash preservation, management must address returns using two separate analytical perspectives:

Perspective A: Category Overages (Product Quality Risk)

Product categories that exceed the 7.20% benchmark—specifically Electronics (8.61%), Books (8.37%), and Clothing (8.19%)—indicate product quality issues. For these categories, an internal operational audit must be launched to investigate potential defective batch manufacturing, supplier packaging vulnerabilities, or product sizing mismatches. These represent hard cash leaks that damage brand equity.

Perspective B: Channel Overages (The Interaction Spin)

Conversely, return rates analyzed by sales channel reveal an elevated 9.14% return rate on Social Commerce. While this represents a financial cost, it reflects customer interaction trends rather than product quality issues. Modern mobile-first digital buyers expect seamless, low-friction, convenient return paths. This elevated rate may simply be a byproduct of frictionless consumer interaction options that build long-term lifetime value, rather than indicating a direct problem with the physical products themselves. Management should focus on optimizing the return processing speed for these channels to lower processing overheads.